Improvements on the schemes VOX and QR–UOV

When minus is a plus

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Abstract. In this article, we present an improvement for QR–UOV and a reparation of VOX. Thanks to the use of the perturbation minus we are able to fully exploit the parameters in order to reduce the public key. It also helps us to repair the scheme VOX. With QR–UOV- we are able to significantly reduce the size of the public key at the cost of a small increase of the signature size. We show that VOX does not bring significant advantage in terms of sizes compared to QR but gives a double security aspect. We also show that the use of minus perturbation is very useful with schemes that uses the QR (Quotient ring) perturbation.

Keywords: Post-Quantum public key signature · Multivariate cryptography · UOV

1 Introduction

In the realm of post-quantum cryptography, multivariate cryptography is one of the main candidates. It uses the difficulty to solve a quadratic system of equation in a finite field (MQ problem). But it requires a structure which makes the equation easier to solve for the signer (in the case of a signature scheme) and hide this structure with a linear change of variable. We can date the beginning of this domain with the introduction of the Matsumoto and Imai scheme C^*. Although the scheme was broken it lead to two new algorithm by Jacques Patarin: Hidden Field Equation (HFE) [Pat96] and Oil and Vinegar (OV) [KS98], the latter later became Unbalanced Oil and Vinegar (UOV) [KPG99a]. For long, HFE was the most studied but after several attacks [BFP13], [BBC+22] it became clear for most that UOV had the most potential. It is in this context that a total of 6 scheme based on UOV were submitted to the recent NIST Post-Quantum Cryptography: Additional Digital Signature Schemes competition. The basic scheme UOV is simple in its principle as the signature only require to solve linear equation.

However, the basic scheme requires large public keys, and many additional technics were added to the scheme in order to limit this problem. For example the Petzoldt technic [Pet20] is used in most UOV-like schemes, as it was proved that it strictly reduced the size of the public key without any drawbacks on the security. Besides, other perturbations were found but no security proof exists for them. Many of the UOV-like schemes submitted to the NIST use this type of technic. Among them we can cite: Mayo [BCC+23], SNOVA [WCD+24], QR–UOV [FIH+24] and VOX [PCF+23]. VOX and QR–UOV have similarities in the sense that they both used the QR (Quotient Ring) technic in order to reduce the public key. However, due to the parameters of VOX, a MinRank attack was found [GD24] that made the authors reconsider their choice of the QR technic. In the end the VOX scheme was not chosen for the second round of the NIST competition. In this article we propose a new set of parameters for VOX with the use of the minus perturbation. We will also reconsider the parameters of QR–UOV with the use of the same perturbation. We will show that this perturbation has two effects. It increases the complexity of some key attacks against both VOX and QR–UOV. It also allows to increase the size of extension of the quotient ring without having to increase the overall amount of equations, allowing for a clear reduction of the public key. We will denote these perturbation as " $-$ " hence we will talk of VOX- and QR–UOV- In this paper we will describe both of the scheme and the perturbation they use. We will update the cryptanalysis with the use of the " $-$ " perturbation. And we will propose a set of parameters for both of them and show the improvement especially on the size of the public key.

1.1 Notations

For our notations, the set of all integers between integers a, b (a and b included) is $\{a \dots b\}$. Row vectors and matrices will be written in **bold**.

A field with q elements is denoted $\mathbb{F}_q$.

Finally we note Tr_n the well known linear mapping trace defined by $\mathrm{Tr}_n : \mathbb{F}_{q^n} \to \mathbb{F}_q, x \mapsto \sum_{i=0}^{n-1} x^{q^i}$

1.2 Formal definition of UOV

For the representation of UOV we will be presenting the reprentation of Beullens [Beu21] and Patarin [KPG99a](both are equivalent in terms of quadratic form).

We present first the representation of Ward Beullens

Let P a quadratic form and U a subspace of F^n such that $n = o + v$ and $\lceil\rangle\Updownarrow(U) = o$ and $P(O) = 0$. Then the public key of UOV consist of a set of quadratic forms $P_1, ..., P_o$ that verify the previous property (the space U is common to all quadratic forms). And the private key is the space O (also called oil space). Let P_i' be the bilinear form associated with the quadratic form P_i. Then the process of signature works in the following way. Say that one want to sign the message Y of the hash of the message Y for commodity we will note both Y. Then the signer will send Y and will have to find X such that $P(X) = Y$. To do so we take a vector $X = o+v$ with $o \in O$ then $\forall i\ P_i(X) = P_i(v)+P_i(o)+P_i'(v, o)$ so $P(o) = 0$ then by fixing v we can obtain the desired solution by solving a linear system in o. Namely: $\forall i\ Y - P_i(v) = P'_i(v,o)$ this system is linear in o hence is easy to solve.

Now we present the representation of Patarin.

Let o, v two integers, $n = o+v$ and q a prime power, and let $\mathbf{x} = (x_1, \dots, x_n)$ be variables over F_q. We call $x_1, \dots, x_v$ the vinegar variables and $x_{v+1}, \dots, x_n$ the oil variables. We call $\mathbf{F} = F_1, \dots F_m : \mathbb{F}_q^n \to \mathbb{F}_q^m$ the central map (in this context $m = o$) where $F_k = \sum_{i=1}^{v} \sum_{j=1}^{n} \alpha_{i,j}^{(k)} x_i x_j$ where $\alpha_{i,j}^{(k)} \in \mathbb{F}_q$. let $\mathbf{F_k}$ the matrix representation of F_k (as in this paper q will not be chosen even, we can uniquely define a matrix representation for the quadratic form F_k by taking the symmetric representation)

Then let $S : \mathbb{F}_q^n \to \mathbb{F}_q^n$ a random linear map and $\mathbf{S}$ it's matrix representation. Then let $\mathbf{P_k} = \mathbf{S}^t \mathbf{F_k} \mathbf{S}$. Then the public key of UOV will be

$$\mathbf{P} = \mathbf{P_1}, \dots, \mathbf{P_m}.$$

1.3 Quotient Ring

In this subsection we will present the use of a Quotient Ring for UOV [FIKT21], and the matrices that are used for QR–UOV and VOX. The goal is to view $N.c \times N.c$ $\mathbb{F}_q$ matrices as $N \times N$ matrices of $\mathbb{F}_{q^c}$.

Let c be a positive integer and $f \in \mathbb{F}_q[x]$ with $degf = c$. Then we for any element g of $\mathbb{F}_q[x]/(f)$, we can define Φ_g^f a $l \times l$ matrix over $\mathbb{F}_q$ such that

$$(1, x, \dots, x^{l-1})\Phi_g^f = (g, xg, \dots, x^{l-1}g).$$

This equation gives:

$$x^{j-1}g = \sum_{i=1}^{c} (\Phi_g^f)_{ij} \cdot x^{i-1} (j \in [c]).$$

This way of defining the matrix Φ_g^f makes the map $g \to \Phi_g^f$ an injective homomorphism from $\mathbb{F}_q[x]/(f)$ to the matrix ring $\mathbb{F}_q^{c\times c}$.

In order to give the full extent of the reversibility of this construction. We introduced as in the notion of the matrix $\mathbf{W} \in \mathbb{F}_q^{c\times c}$ such that $\mathbf{W}\Phi_g^f$ is symmetrical.

Theorem 1. *Let $f \in \mathbb{F}_q[x]$ with $degf = c$. Then there exists an invertible matrix $\mathbf{W} \in \mathbb{F}_q^{c\times c}$ such that for all $g \in \mathbb{F}_q[x]/(f)$, $\mathbf{W}\Phi_g^f$ is a symmetric matrix.*

A proof of the theorem is given in. So the QR transformation allows to consider a $UOV(q, v, o, m)$as a $UOV(q^c, v/c, o/c, m)$ hence we will note $V = v/c$ and $O = o/c$. This construction allows to stock the public key of UOV as a matrix of elements in $\mathbb{F}_{q^c}$. Every entries of the matrix can be represented by only c elements, hence reducing the size of the public key. We will later give a formula computing the size of the public depending on the parameters.

1.4 Plus hat perturbation

The Plus hat perturbation (or $\hat{+}$) is only used in VOX [FmRPP22]. It involves adding random quadratic forms to the central map. Recall the previous notations of UOV (namely: o, v, m, n, q).

Let t be an integer such that $t \leq n$ then let the central map of $UOV\hat{+}$, $F = F_1, \dots F_m$ The $m - t$ last polynomials are defined like in UOV : $F_k = \sum_{i=1}^{v} \sum_{j=1}^{n} \alpha_{i,j}^{(k)} x_i x_j$.

However the first t polynomials are random quadratic polynomials. Like in UOV we define $\mathbf{F_k}$ as the matrix representation of F_k. And let $\mathbf{S}$ be a $n \times n$ matrix of $\mathbb{F}_q$. We define another matrix $\mathbf{T} \in \mathbb{F}_q^{n \times n}$. We define the public key in the following way

$$\mathbf{P} = \mathbf{P_1}, \dots \mathbf{P_n} = (\mathbf{S}^t \mathbf{F_1} \mathbf{S}, \dots, \mathbf{S}^t \mathbf{F_m} \mathbf{S})\mathbf{T}$$

In order to sign, we have to do an exhaustive search on all the values of the t random polynomial, effectively costing q^t times the number of signature of a regular UOV (Complete signature protocol will be explained later on). VOX is then a scheme QR–UOV $\hat{+}$.

2 Cryptanalysis

In this section we will make an overview of the attacks used on UOV and their specific effect on the schemes QR–UOV and VOX.

2.1 Direct attack

In this subsection, we will describe the Direct attack [BWP05] [FP09](sometimes called by abuse Gröbner attack). The goal of this attack is to forge a signature by finding a solution in $x \in \mathbb{F}_q^n$ of the equation $P(x) = y$ for a given public key P and vector $y \in \mathbb{F}_q^m$. In a standard UOV we have that $m \leq n$, then as in let $\alpha = \lfloor \frac{n}{m} \rfloor - 1$ and reduce $(n - m + \alpha)$ variables and α equations thus we obtain a quadratic system with $m - \alpha$ equations and variables. we can then study the standard overdetermined case. The current best method is called the hybrid method, which consist in randomly guesses with $0 \leq k \leq n$ then applying an algorithm like F4 [Fau99], F5 [Fau02], or XL [CKPS00]. The complexity evaluation is

$$\min_k \left(O(q^k 3 \binom{n-k+2}{2} \binom{d_{reg}+n-k}{d_{reg}}^2 \right).$$

d_{reg} is the degree of regularity of the system. For semi-regular system the degree of regularity is the degree of the first non positive term in the series:

$$\frac{(1-z^2)^m}{(1-z)^{n-k+1}}.$$

Although the result is not proven we usually suppose that the public key of UOV is semi-regular (the public key of QR–UOV and VOX are also considered semi-regular). For NIST specification the authors of UOV like scheme usually have chosen parameters (or at least tried to chose parameters) such that the best attack is the direct attack.

With this attack we must consider two cases. One in the big field where we consider a system with m equations, and $O + V$ variables in $\mathbb{F}_{q^c}$ and another system with m equations $o + v = c * (O + V)$ variables in $\mathbb{F}_q$

2.2 Key Recovery Attacks on UOV

We will describe here some of the method used to recover the private key. It is important to note that due to the result of Pébereau in [Péb24], we only need to recover one vector that belongs to the oil vector space to recover the whole oil space. Hence effectively recovering the private key. However recovering that said vector remains a difficult problem.

Note that the QR perturbation induces two key recovery problems. Indeed, the attacker can either look at as public key of a QR–UOV (and by extent VOX) as a:

- $UOV(q^c, V, O, m)$,
- $UOV(q, v, m, m)$,

Here, $UOV(q, v, m, m)$ represents the UOV with v vinegar variables and o oil variables (here $m = o$). And $UOV(q^c, V, O, m)$ represents the UOV in the QR extension.

Kipnis Shamir Attack It's historically the attack that imposed OV (Oil and Vinegar) to become UOV (Unbalanced Oil and Vinegar) [KS98].

Let $\mathbf{W}_i, \mathbf{W}_j$ two invertible matrices from the set of linear combinations of the representation matrices $\mathbf{P}_1, \ldots, \mathbf{P}_m$ of the public key. The goal is to find vectors of the oil space by computing $\mathbf{W}_i^{-1}\mathbf{W}_j$.

The complexity of the attack is estimated as:

$$O(q^{v-o-1}.o^4).$$

This attack also applies on VOX (or in this case $\hat{+}$) but it requires first to cancel the t full quadratic forms and it requires on average to combine q^{2t} equations hence the complexity will be estimated as:

$$O(q^{2t}.q^{v-o-1}.o^4).$$

Intersection Attack This attack was found by Beullens in, In the case of $v < 2o$ for an integer $k \geq 2$ such that $k < \frac{v}{v-o}$, let $\mathbf{L}_1, \ldots, \mathbf{L}_k$ be k invertible matrices randomly chosen from a set of linear combination of the public key matrices $\mathbf{P}_1, \ldots, \mathbf{P}_m$.

Then in order to recover a vector from the oil space. We solve the following equations for $y \in \mathbb{F}_q^n$:

$$\begin{cases} (\mathbf{L}_j^{-1}y)^\top \mathbf{P}_i(\mathbf{L}_j^{-1}y) = 0 \quad \text{for } (i \in [m], 1 \leq j \leq k) \\ (\mathbf{L}_j^{-1}y)^\top \mathbf{P}_i(\mathbf{L}_l^{-1}y) = 0 \text{ for } (i \in [m], 1 \leq j < l \leq k) \end{cases}$$

A solution z of the system is not directly a vector in the oil space but $\mathbf{L}_j^{-1}z$ is a vector of that space. The space of solutions has $ko-(k-1)v$ dimensions. It implies solving a quadratic system with $n-(ko-(k-1)v)$ variables and $\binom{k+1}{2}m - 2\binom{k}{2}$ equations. The value of k is chosen to minimize the complexity of the attack. The complexity against $UOV(q^c, V, O, m)$ is given by the formula:

$$\min_k \left(O\left(q^{v-2m+c} q^{ck} 3 \binom{n/c-k+2}{2} \binom{d_{reg}+n/c-k}{d_{reg}}^2 \right) \right)$$

and against $UOV(q, v, m, m)$:

$$\min_k \left(O\left(q^{v-2m+1} q^{k} 3 \binom{n-k+2}{2} \binom{d_{reg}+n-k}{d_{reg}}^2 \right) \right)$$

As in the previous section d_{reg} denotes the degree of regularity previously defined. It is due to the fact that the resolution of the equations involve the use of MQ solver.

It should be noted that in the case that $v \geq 2o$ the intersection attack is now probabilistic with a probability of success of $q^{-v+2o-1}$. Hence the complexity is now q^{v-2o+1} times the complexities we previously mentioned.

Furthermore, the use of $\hat{+}$ does impact the complexity as like the Kipnis-Shamir attack it seems that it is required to cancel the t full quadratic forms hence it requires q^{2t} operation times the complexity of the attack.

Reconciliation Attack The main idea of the reconciliation attack [DYC+08] is to treat a vector $\mathbf{y}$ of $\mathbf{S}^{-1}O$ as variables we then have to solve the system $\mathbf{y}^T\mathbf{P_i}\mathbf{y} = 0$ the dimension of $\mathbf{S}^{-1}O$ is o so we then solve a system of m equations in $n-o$ variables. Two cases arises: if $v < m$ and $v > m$. Most of the time in UOV like system we have $v > m$ Therefore, in order to determine a unique solution we are using the linear stability of the y variables in order to obtain more solutions namely: Let $\mathbf{y}_1, \ldots, \mathbf{y}_k$ be vectors of $\mathbf{S}^{-1}O$. We obtain the following system:

$$\begin{cases} \mathbf{y}_j^T \mathbf{P_i} \mathbf{y}_j = 0 (1 \leq i \leq m, 1 \leq j \leq k), \\ \mathbf{y}_j^T \mathbf{P_i} \mathbf{y}_l = 0 (1 \leq i \leq m, 1 \leq j < l \leq k), \end{cases}$$

It means that a lower bound of the complexity in that case is the same as to solve a quadratic system with v variables and equations.

In the case of $v < m$ we estimate the complexity of the reconciliation attack to be equivalent as of solving a quadratic system of m equations in v variables.

Again with QR–UOV we can either attack on the small or the big field but our experience showed that we obtain the lowest complexity when we consider the attack on the big field. In that case the complexity is:

$$\min_k \left(\mathbf{O} \left(q^{ck} \cdot 3 \cdot \binom{V-k+2}{2} \cdot \binom{d_{reg}+V-k}{d_{reg}}^2 \right) \right),$$

with $k \leq V$.

Rectangular MinRank Attack The rectangular attack was first proposed by Beullens [Beu22]. And Furue and Ikematsu [FI23] showed that the attack can work against $UOV(q^c, V, O, m)$. So in the rest of the section we will only consider UOV in the QR extension. Here the attack slightly differ when we consider $\hat{+}$ because we need to consider the second matrix $\mathbf{T}$, that does not exist in the general case. We will then adopt a general approach where we consider that in every cases $\mathbf{T}$ is present but eventually equal to the identity matrix that way we will not have to differentiate the cases where $\hat{+}$ is present or not. Recall that $N = O + V = n/c$.

We will describe the transformation $\tilde{\mathbf{G}}$. Let $\{\mathbf{G}_1, \ldots, \mathbf{G_m}\}$ be a set of matrices of size $n \times n$. Then let the set $\{\tilde{\mathbf{G}}_1, \ldots, \tilde{\mathbf{G}}_\mathbf{n}\}$ of $n \times m$ matrices be defined as followed.

$$\tilde{\mathbf{G}}_\mathbf{k} = (g_1^{(k)}, \ldots, g_m^{(k)})$$

where $g_i^{(j)}$ is the j-th column vector of the matrix $\mathbf{G}_i$. Then the equation

$$\mathbf{P}_1, \ldots, \mathbf{P}_m = (\mathbf{S}^t \mathbf{F_1} \mathbf{S}, \ldots, \mathbf{S}^t \mathbf{F_m} \mathbf{S})\mathbf{T}$$

becomes

$$\tilde{\mathbf{P}}_1, \ldots, \tilde{\mathbf{P}}_N = (\mathbf{S}^t \tilde{\mathbf{F}}_\mathbf{1} \mathbf{T}, \ldots, \mathbf{S}^t \tilde{\mathbf{F}}_\mathbf{N} \mathbf{T})\mathbf{S}.$$

(note that we consider the matrix $\mathbf{F}_i$ as matrices with coefficient in $\mathbb{F}_{q^c}$, hence they are of size $N \times N$) We focus on the rank of $\mathbf{S}^t \tilde{\mathbf{F}}_\mathbf{N} \mathbf{T}$. We know that the first t (here t is eventually 0) columns of $\tilde{F}_N$ are random (due to the $\hat{+}$ perturbation). Hence the rank of the matrix $\tilde{\mathbf{F}}_\mathbf{N}$ is $V + t$.

It implies that there exist a linear combination of the matrices $\tilde{\mathbf{P}}_1, \ldots, \tilde{\mathbf{P}}_N$ such that their rank is $V + t$.

Then we can use the support minor method to solve the MinRank problem. We will not use the method used by Ding *et al.* have found [GJP+24] it is only relevant because the parameters of VOX specification used to have $t \geq O$ which required another type of method to solve. However in our case we will always have $t \leq O$.

To give a complexity evaluation of this attack we are using the analysis of [Beu22] updated for QR–UOV in [FIH+24]. We have also used the complexity analysis of [PCF+23] for the effect of the $\hat{+}$ perturbation.

In their analysis they have found that the complexity of the MinRank attack is:

$$3\binom{m'}{V+t}^2 \binom{V+t+b_{min}-1}{b_{min}}^2 (V+t+1)^2$$

where m' is the optimal integer between $V+t+1$ and m and b_{min} is evaluated in the following way:

let

$$R'(b) = \sum_{i=1}^{b} (-1)^{i+1} \binom{m'}{V+t+1} \binom{N+i-1}{i} \binom{V+t+1+b-i}{b-i}$$

and let

$$G(t_1,t_2) = (1-t_1^2)^m \cdot \left(\binom{m'}{V+t} t_2 + \sum_{b=1}^{V+t+1} (\binom{m'}{V+t} \binom{V+t+b-1}{b} - R'(b) \right) t_1^b t_2)$$

and let $G(t_1,t_2)_{b,1}$ be the coefficient of $t_1^b t_2$ then $b_{min} = min\{b, G(t_1,t_2)_{b,1} \leq 1\}$. We may notice that if $t = 0$ we find the regular QR–UOV complexity for the attack. Furthermore, a small improvement on this attack have been found in [FI23], it allows a range of choice for the target rank and it has been taken into account in our security analysis.

3 Minus for UOV like schemes

In this section we will describe the perturbation we want to use for our new schemes and the effect on the two main attacks (in terms of effectiveness) on VOX and QR–UOV.

3.1 Minus perturbation

The minus perturbation is a very simple perturbation and has been introduced by Goubin *et al.* in his introductive article for UOV [KPG99b] (in this article it was proposed for another algorithm HFE but the principle remains the same). It simply involve hiding some of the public key polynomial. The goal by hiding some of the polynomial is to hide part of the structure of UOV.

Let $\mathbf{P} = (\mathbf{P}_1, \ldots, \mathbf{P}_m)$ be the public key of a UOV like scheme (e.g VOX, QR–UOV, UOV...) then the perturbation minus with parameter a gives a new public key $\mathbf{P}' = (\mathbf{P}_1, \ldots, \mathbf{P}_{m-a})$.

We can remark that for a fixed $x \in \mathbb{F}_q^n$, then $[\mathbf{P}x]_{m-a} = \mathbf{P}'x$, by $[]_{m-a}$ we mean that we only select the $m-a$ first values. Hence, in order to sign, the signer that knows the private key hence knows $\mathbf{P}'$ and $\mathbf{P}$ can sign like if it was a UOV without any perturbation by adding random values to $y_{m-a+1}, \ldots, y_m$.

Signature Protocol We give here a detailed version of the signature protocol for UOV with minus perturbation

- The verifier compute $y = \mathrm{Hash}(\mu, r) \in \mathbb{F}_q^{m-a}$
- The signer generates at random $(y_{m-a+1}, \ldots, y_m) \in \mathbb{F}_q^a$

- Then the signer finds x such that $x^t\mathbf{S}^t\mathbf{F}\mathbf{S}x = y||(y_{m-a+1},\dots,y_m)$ to do so we find x' such that$x'^t\mathbf{F}x' = y||(y_{m-a+1},\dots,y_m)$.
- The signer uses the structure of $\mathbf{F}$ and fixes the vinegar variables and proceed to a Gaussian reduction. if it doesn't work the signer regenerate $(y_{m-a+1},\dots,y_m)$ and tries to invert again.
- The Signer send $x = S^-1x'$.

Essentially the signature protocol does not change much from a regular UOV. As the signer can ignore the perturbation and simulate the signature of a regular UOV.

3.2 Effect on attacks

We will now describe the effects of the minus perturbation on the attacks we previously described. As we did in the previous section we can differentiate two types of attacks.

Direct attacks The direct attack, ignores the structure of UOV and rather tries to directly invert the system like a regular quadratic system. Hence the effectiveness of this attack relies only on the number of variables, equations and field size rather the number of oil and vinegar variables. We must distinct to cases, the effect of minus on the big field and on the small field.

On the small field Minus will decrease the complexity of this attacks. In a greatly underdetermined case, decreasing the number of equations while maintaining the number of variables and the size of the field will lead to a decrease of the complexity. Indeed, the complexity of the attacks is directly related to the degree of regularity d_{reg} and it decreases as the number of equations decreases.

On the big field it's the opposite effect. Indeed the number of variables will be lower than the number of equations. In that case having a lower number of equations with the same number of variables will increase the complexity of the direct attack.

Kipnis Shamir and Intersection attacks Both of the attacks rely on the structure of UOV, as they try to exploit linear combinations of the public key to find oil vectors. Furthermore, the complexity of the Kipnis Shamir attack does not involve the number of equations but the size of the oil space. Hence the complexity remains unchanged.

$$q^{v-o-1} \cdot o^4.$$

The intersection attack is slightly changed as the number of equations is slightly lower while the number of variables stays the same. In other words the complexity still is:

$$min_k(O(q^{v-2m+1}q^k 3\binom{n-k+2}{2}\binom{d_{reg}+n-k}{d_{reg}}^2))$$

but the change lies in d_{reg} as the number of equations is lower. It becomes $3(o-a)$ equations instead of $3o$.

Reconciliation attacks The reconciliation attack is impacted by the minus perturbation. As we stated the best case for the reconciliation attack over QR–UOV is on the big field. And in that case we will have $V < m$. Furthermore, we have indeed experienced that lowering the number of equations will tend increase the complexity of the resolution because the number of variables is lower than the number of equations thus the closer we get from the case $m = n$ which is largely believed to be the worst case for the MQ problem [TW12], [BFP10], [BFS05]. Indeed, the more we add equations to an overdetermined system the easier it is to solve. So on the contrary when we have an overdetermined system the more equation we discard the harder it is to solve up until $m = n$.

After that if we keep discarding equations then the complexity will go down. Which means that at this point the effect of the minus perturbation will be detrimental to the security of the system. However, we only discard a limited number of equations such that we are never reaching the case $m = n$. So overall in our parameters the minus perturbation helps increasing the complexity of this attack.

Rectangular attacks The rectangular attack is very impacted by the diminution of the number of equations. Indeed we must recall that the goal is to find a rank defect from a linear combination of equation taken from the public key. Recall that:

$$(\tilde{\mathbf{P}}_1, \dots, \tilde{\mathbf{P}}_n) = (\mathbf{S}^t\tilde{\mathbf{F}}_\mathbf{1}\mathbf{T}, \dots, \mathbf{S}^t\tilde{\mathbf{F}}_\mathbf{n}\mathbf{T})\mathbf{S},$$

and that the matrices are now of size $N \times m$ hence a diminution of m implies a diminution of the probability to have found a rank defect (but the rank defect stays the same as $m \gg N$). So intuitively it increases the complexity of the attack.

Furthermore: Recall that b_{min}

$$G(t_1, t_2) = (1 - t_1^2)^m \cdot \left(\binom{m'}{V+t} t_2 + \sum_{b=1}^{V+t+1} \left(\binom{m'}{V+t} \binom{V+t+b-1}{b} - R'(b) \right) t_1^b t_2 \right)$$

is the first negative coefficient of the form $t_1^b t_2$ but if m decreases then the negative coefficient of $(1 - t_1^2)^m$ will decrease and thus b_{min} will increase.

3.3 Symmetric Algebra Attack

A recent attack was found using symmetric algebra [JPH+25]. It generalizes the attack of Lars Ran [Ran25] that could only be used against field with characteristic 2. Due to the utilization of the QR variant, our Fields always have an odd characteristic therefore we need to use the generalization of the attack [JPH+25].

In this article the authors have applied their attack against QR–UOV but they found out that their attack was less effective than the reconciliation attack due to the large characteristic of the Field. We have kept the same characteristic in our parameters set, and found the same result. Therefore, we will not further describe this attack.

3.4 Advantages

We will further describe the advantages that we can obtain from the perturbation minus. We show that the perturbation can be a good addition to some variants of UOV.

Security Our study show that minus is effective against certain attacks, especially rectangular attack, reconciliation and the direct attack on the big field. It means that an addition of some minus perturbation can help to defend certain types of algorithm vulnerable to this attack. In particular we have in mind VOX. On the other hand it should be noted that removing some equations of the public key has a negative impact on the direct attack on the small field. Indeed we show that direct attack on the small field are more effective against an underdetermined system (UOV like schemes) with less equations (with the same number of variables and field sizes). Hence, the number a of removed equations should be limited.

Public Key Nonetheless, it is quite obvious that with a fixed number of variables $n = o + v$, the public key will decrease as a is increased. Indeed, less matrices are required to represent the public key. This leads to the conclusion that having a limited number of removed equations can strictly improve some parameters of schemes that took a bit of margin against direct attacks.

3.5 Signature

The attack on the big field will tend to impose a large number of vinegar variables that will in turn increase the size of the signature. On the other hand due to the effect of the minus perturbations. We are allowed to have less vinegar variables. Hence, limit the size of the signature. Which is especially interesting in the case of large QR extension.

QR and Minus At this point, we could note that lowering the number of oil variables would have a similar effect as getting rid of some equations. And lowering the number of oil variables is even more effective if the goal is to lower the public key. In fact in the NIST competion only PROV [GCF+23] retained this in order to obtain a formal proof for their algorithm. However, our main point is that in relation with QR it makes a lot of sense to use the minus perturbation.

As we noted the key aspects of direct attacks is the number of equations, the number of variables and the field size. However, when one uses the QR variant it impose the oil field and hence the number of equation to be a multiple of the extension coefficient i.e $m = O \cdot c$. But, it means that in some cases we do not need this much equations to avoid direct attacks. For example, let $v = 406, o = 70, c = 14, q = 31$ and let $\lambda = 143$ be the security level. In this case the direct attack is expected to invert the system with a complexity of 173.9. However, we have a little margin and naturally in order to optimize the public key we could try to diminish the size of the oil space. But if we do so the expected complexity is now 129.9. So we are now below the security level. Then, the perturbation minus allows for more agency. We can limit the number of equations to the strict minimum while use the QR variant to its full potential. It means we can increase the QR coefficient c and gain a net diminution of the public key. Indeed, if c is too high then we have no margin for the choice of o, and it can lead to numerous "useless" equations. It means that sometimes it is preferable to limit the number c as it does not give a significant diminution of the public key. The use of minus perturbation allows us to increase the c coefficient while preserving the number of equations which leads to a reduction of the size of the public key.

In the following section we will give a new sets of parameters for VOX- and QR–UOV- that will illustrate the improvement we can obtain in comparison to QR–UOV.

4 Design Rationale and Parameters

We will describe the new schemes VOX- and QR–UOV-. We will add to the original schemes the perturbation minus. It means that the public key will have less equation than the size of the oil space. The scheme VOX- is a UOV scheme with the addition of QR, $\hat{+}$ and minus perturbation.

So let v, o, q, c, t, a be its parameters where v is the number of vinegar variables, o the number of oil variables, q the size of the field, c the QR extension, t the number of random polynomials and a the number of equations that we hide (i.e the number of equations is $m = o - a$). We note $n = v + o$, $V = v/c$, $O = o/c$, $N = n/c$ and $A = \{\Phi_g^f \in \mathbb{F}_q^{c\times c} | g \in \mathbb{F}_q[x]/(f)\}$ where f is an irreducible polynomial of $\mathbb{F}_q[x]$ of degree c and Φ as defined in section 1.2. We also define $A^{N\times N}$ the set of matrices of size N with coefficient in A. We choose $\mathbf{F} = (\mathbf{F}_1, \dots, \mathbf{F}_o)$ a set symmetric matrices in $\mathbf{W}A^{N\times N}$. The t first $\mathbf{F}_i$ are chosen randomly and the next $o - t$ matrices are chosen such that the lower $O \times O$ block is zero, i.e: $\begin{pmatrix} M & L \\ G & 0 \end{pmatrix}$. Then choose an invertible matrix $\mathbf{S} \in A^{N\times N}$ and the invertible matrix $\mathbf{T} \in \mathbb{F}_q^{N\times N}$.

We define

$$\mathbf{P} = (\mathbf{P}_1, \dots, \mathbf{P}_o) = (\mathbf{S}^t\mathbf{F}_1\mathbf{S}, \dots, \mathbf{S}^t\mathbf{F}_o\mathbf{S})\mathbf{T}$$

then the public key is $\mathbf{P'} = (\mathbf{P'}_1, \dots, \mathbf{P'}_m) = (\mathbf{P}_1, \dots, \mathbf{P}_{o-a})$. The private key is $(\mathbf{F}, \mathbf{S}, \mathbf{T})$

The design rationale of QR–UOV- is essentially the same with $t = 0$ so we will not re-describe it.

N.B. For the computation of the signature, public key and private key sizes we consider that we are using the Petzoldt [Pet20] technique although we never formally described it as it does not bring significant changes to the logic of the design.

Signature Protocol for VOX-

- The verifier computes $y = \mathrm{Hash}(\mu, r) \in \mathbb{F}_q^{m-a}$ where μ is a hash value of a public seed $seed_{pk}$ and a message M, r is a salt.
- The signer generates at random $(y_{m-a+1}, \ldots, y_m) \in \mathbb{F}_q^a$
- Then the signer needs to find x such that $x^t\mathbf{S}^t\mathbf{F}\mathbf{S}x = y||(y_{m-a+1}, \ldots, y_m)$.
- To do so we find x' such that $x'^t\mathbf{F}x' = y||(y_{m-a+1}, \ldots, y_m)$.
- First the signer generates $\tilde{\mathbf{F}}$ that corresponds to the central map with coefficients in $\mathbb{F}_q$.
- The signer fixes the vinegar variables. He obtains two systems with t quadratic equations and $o - t$ affine equations.
- The signer eliminates $o-t$ variables with the affine equations, in other words he expresses $o-t$ variables as affine combination of t variables (e.g $o-t$ first are expressed in function of the t last)
- He solves the quadratic system of t equations in t variables with a Gröbner basis algorithm.
- If the previous step doesn't work the signer regenerate $(y_{m-a+1}, \ldots, y_m)$ and tries to invert again.
- The signer obtains the values of t variables then he can deduce $o-t$ variables with the help of the affine equations. He can then recover x'.
- The Signer sends $x = S^{-1}x'$.

Signature Protocol for QR–UOV-

- The verifier compute $y = Hash(\mu, r) \in \mathbb{F}_q^{m-a}$
- The signer generates at random $(y_{m-a+1}, \ldots, y_m) \in \mathbb{F}_q^a$
- Then the signer needs to find x such that $x^t\mathbf{S}^t\mathbf{F}\mathbf{S}x = y||(y_{m-a+1}, \ldots, y_m)$ to do so we find x' such that $x'^t\mathbf{F}x' = y||(y_{m-a+1}, \ldots, y_m)$.
- First the signer generates $\mathbf{F}'$ that corresponds to the central map with coefficient in $\mathbb{F}_q$.
- the signer fixes the vinegar variables the solve then solve the induced affine system of o equations in o variables.
- If the previous step doesn't work the signer regenerate $(y_{m-a+1}, \ldots, y_m)$ and tries to invert again.
- x' is then obtained by the fixed variables and the o variables obtained from the resolution of the affine system.
- The Signer sends $x = S^{-1}x'$.

4.1 Parameters

In this section we give propositions of parameters that corresponds to the NIST levels of security. For VOX- we give new parameters that are much different from the ones submitted to the first round of competition in their specification 1.0.a. In fact we were unable to obtain parameters such that the $\hat{+}$ perturbation allowed for better parameters that QR–UOV-. Indeed, both of the direct attacks (on which $\hat{+}$ have no effect) impose a high amount of vinegar variables. This makes all key recovery attacks impossible. However, having a small number of $\hat{+}$ most certainly increased the complexity of key recovery attacks. It means that in the case of a potential new key recovery attack we are likely to be more secure.

On the other hand for QR–UOV- we try to increase slightly the size of c. We then use the coefficient a to maintain the number of equations as low as possible in order to limit the size of the public key.

o,v,O,V,q,c,a	—PK—	—Sig—	Sec level
55,245,11,49,127,5,3	15022	294	I
72,696 ,6,58,31,12,4	10643	507	I
80,375,16,75,127,5,4	45194	445	III
96,1008,8,84,31,12,2	25195	731	III
108,612,18,102,127,6,4	93279	692	V
20,1635,8,109,31,15,2	39524	1150	V

Table 1. Parameters and sizes (Byte) for QR–UOV-

o,v,O,V,q,c,a	—PK—	—Sig—	Sec level
54,156,18,52,127,3,0	24255	215	I
70,600,7,60,31,10,0	12266	435	I
78,228,29,76,127,3,0	71891	292	III
100,890,10,89,31,10,0	34399	643	III
38,108,114,324,127,3,0	173676	392	V
120,1120,12,112,31,10,0	58532	807	V

Table 2. Parameters and sizes (Byte) for QR–UOV

As we can see by comparing both table 1 and 2 that we obtain significant diminution to the public keys. Our goal was to keep $o = c \times O$ and $v = c \times V$ as close a possible from the original version of QR–UOV. We kept the same field size. But we increased the QR coefficient c. We obtain a slight increase of the signature. This is due to the fact that we have to slightly increase the number of variables because of the augmentation of the c coefficient. It should be noted that without the addition of the minus perturbation, the size of the public key would have been comparable to the values of the original QR–UOV. In our case we obtain a significant difference due to both perturbation.

We can compare the sizes of both VOX- and QR–UOV-. Overall the signatures we can obtain with VOX- is better than the one we obtain with QR–UOV-. We can achieve that without having significant differences in terms of public key sizes. However VOX- is expected to be slower than QR–UOV- as the use of $\hat{+}$ is costly.

We can also observe that the sizes of QR–UOV- are significantly better than the one of QR–UOV. This is due to both the decision to increase c and the use

o,v,O,V,q,c,a,t	—PK—	—Sig—	Sec level
55,245,11,49,127,5,3,2	15022	294	I
72,696 ,6,58,31,12,4,2	10643	507	I
80,375,16,75,127,5,4,4	45194	445	III
96,1008,8,84,31,12,2,4	25195	731	III
108,612,18,102,127,6,4,6	93279	692	V
120,1635,8,109,31,15,2,6	39524	1150	V

Table 3. Parameters and sizes (Byte) for VOX-

o,v,O,V,q,c,a,t	—PK—	—Sig—	Sec level
52,65,4,5,251,13,0,6	6776	133	I
54,63,6,7,251,9,0,6	10185	133	I
75,90,5,6,1021,15,0,7	21108	189.0	III
77,88,7,8,1021,11,0,7	29656	189	III
102,119,6,7,4093,17,0,8	54648	253	V
104,117,8,9,4093,13,0,8	73033	253	V

Table 4. Parameters and sizes (Byte) for VOX

of the minus perturbation. Without the latter increasing c would not have been as effective.

For VOX we could not obtain better parameters, hence we obtain worse results for the public key. But it should be noted that the parameters of VOX were highly vulnerable. Hence, the comparison is somewhat unpractical.

4.2 Security analysis

We will directly give the complexity formula of each attacks and then give a table that gives the complexity for each sets of parameters. We bring in the table below a recap of the complexity formulas for each attacks for QR–UOV- and VOX-.

Attack	QR–UOV-	VOX-
Direct attack	$\min_k(O(q^k 3\binom{n-k+2}{2}\binom{d_{reg}+n-k}{d_{reg}}^2)$	$\min_k(O(q^k 3\binom{n-k+2}{2}\binom{d_{reg}+n-k}{d_{reg}}^2)$
Kipnis-Shamir	$q^{v-o}o^4$	$q^{2t}q^{v-o-1}o^4$
Intersection	$\min_k(O(q^{v-2o+1}q^k 3\binom{n-k+2}{2}\binom{d_{reg}+n-k}{d_{reg}}^2))$	$\min_k(O(q^{2t}q^{v-2o+1}q^k 3\binom{n-k+2}{2}\binom{d_{reg}+n-k}{d_{reg}}^2))$
Rectangular attack	$\binom{lV+d(lV)+D_{mgd}}{D_{mgd}}^\omega$	$\binom{lV+d(lV+t)+D_{mgd}}{D_{mgd}}^\omega$

Table 5. Complexity formula for each attacks for QR–UOV- and VOX-

We need to recall that d_{reg} and D_{mgd} are the total degree of the first non positive monomial in the respective following series:

$$\frac{(1-z^2)^m}{(1-z)^{n-k+1}}$$

and

$$\frac{\Pi_{i=1}^{k}(1+t_0t_i)^m}{(1-t_0)^{lV}(1-t_1)^r\dots(1-t_k)^r}.$$

Where $V = v/c$.

We then give the complexity of each attack for each set of parameters for VOX- and QR–UOV-

o,v,O,V,q,c,a	Direct	Reconciliation Attack	Rect	sec lev
55,245,11,49,127,5,3	145	148	145	I
72,696 ,6,58,31,12,4	142	145	155	I
80,375,16,75,127,5,4	207	261	210	III
96,1008,8,84,31,12,2	211	207	214	III
108,612,18,102,127,6,4	273	304	280	V
120,1635,8,109,31,15,2	271	272	277	V

Table 6. Complexity for each attack and each set of parameters for QR–UOV- in bit

o,v,O,V,q,c,a,t	Direct	Reconciliation Attack	Rect	sec lev
55,245,11,49,127,5,3,2	145	148	145	I
72,696 ,6,58,31,12,4,2	142	145	155	I
80,375,16,75,127,5,4,4	207	261	210	III
96,1008,8,84,31,12,2,4	211	207	214	III
108,612,18,102,127,6,4,6	273	304	280	V
120,1635,8,109,31,15,2,6	271	272	277	V

Table 7. Complexity for each attack and each set of parameters for VOX- in bit

We obtained an evaluation of the complexity of Direct Attack, and reconciliation attack with the help of the cryptographic estimators [EVZB24]. We computed “manually” the complexity of the Rectangular attack. We can observe that most of the time the direct attack is the best attack in both cases. For certain parameters the rectangular attack is also close from the security level. For example the two first set of parameters give a complexity of the direct attack 2^{145} and 2^{142} (resp) and 2^{145} and 2^{155} for the rectangular attack.

These numbers are so close that it means that without the minus perturbation we would not even have been able to obtain such parameters. The main way to counter the rectangular attack is to increase the vinegar variables. It only increases slightly the complexity of the direct attack while being very effective against the Rectangular attack. But it implies to increase the signature size and if the number of vinegar variables is too high one must increase the size of the oil space which in turn implies an rise of the public key. All in all the flexibility that the minus gives us makes these parameters possible.

5 Conclusion

In this paper, we presented an improvement for UOV-like schemes that uses the QR variant. We showed that we could obtain similar security properties with better public key sizes. Indeed, public keys of multivariate schemes are one of the main issues hence, we think that this improvement is worth implementing for QR–UOV. It allows for more flexibility in terms of parameters without any clear drawbacks. It allowed us to consider higher degrees of extension that in turn directly induced a reduction of the public key. We also were sometimes able to reduce the size of the vinegar space because of the addition of the minus perturbation that in turn allowed to also reduce the signature size. We managed to gain significant improvement of the public key in all cases. Those improvement would have been much more difficult to obtain without the addition of the minus perturbation.

We also used this technique to improve VOX in order to give new parameters. Our goal was to keep the QR variant as it gave good results in terms of public key. We showed the need of the minus variant as both Rectangular and Direct attack have very similar performances. The decision of the authors of VOX to get rid of the QR variant is explained by the lack of flexibility it induces upon the parameters (o should be a multiple of c) and the fact that the $\hat{+}$ perturbation fails to compensate. We were not able to use the $\hat{+}$ to improve key sizes, but we think that the perturbation is interesting for pure security purposes against key recovery attacks. An improvement of the article would be to implement these variants in order to make experiments about their performances. We might expect QR little differences as we tried to keep our parameters as close as possible from the original (we kept the same q and o close, those are the parameters that have the main impact on the complexity of the signing process).

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